

$$(a) \vec{F} = [z^3, x^3, 3y]$$

$$\text{curl } \vec{F} = [3-0, 3z^2-0, 3x^2-0]$$

$$= [3, 3z^2, 3x^2]$$

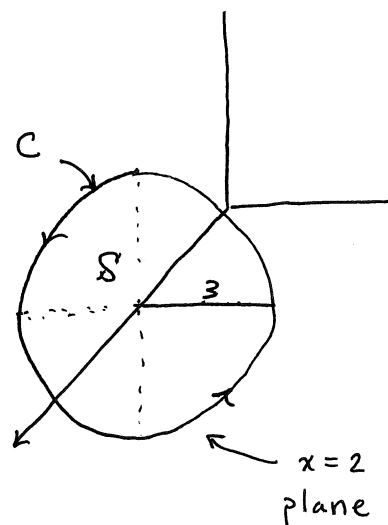
Since I know the area of S and

$\text{curl } \vec{F} \cdot \vec{n}$ is constant, then by Stokes' Thm:

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_S \text{curl } \vec{F} \cdot \vec{n} dA$$

$$= \iint_S 3 dA = 3 \cdot \text{Area}(S) = 3(\pi(9))$$

$$= 27\pi$$



$$\vec{n} = [1, 0, 0]$$

$$(b) \vec{F} = [z^3, x^3, y^2]$$

$$\text{curl } \vec{F} = [2y, 3z^2, 3x^2]$$

In this case, $\text{curl } \vec{F} \cdot \vec{n}$ is not constant
So let's parametrize S

$$\begin{array}{ll} x=2 & u=r \\ y=r \cos \theta & \longrightarrow \vec{r}(u,v) = [2, u \cos v, u \sin v] \\ z=r \sin \theta & v=\theta \end{array} \quad \left. \begin{array}{l} 0 \leq u \leq 3 \\ 0 \leq v \leq 2\pi \end{array} \right\} \begin{array}{l} \text{Region} \\ R \end{array}$$

$$\vec{r}_u = [0, \cos v, \sin v]$$

$$\vec{r}_v = [0, -u \sin v, u \cos v]$$

$$\Rightarrow \vec{N} = \vec{r}_u \times \vec{r}_v = [u \cos^2 v + u \sin^2 v, 0, 0]$$

$$= [u, 0, 0]$$

By Stokes' Thm:

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R \text{curl } \vec{F}(\vec{r}) \cdot \vec{N} du dv$$

$$= \int_0^{2\pi} \int_0^3 2u^2 \cos v du dv$$

$$= \left(\int_0^{2\pi} \cos v dv \right) \cdot \left(\int_0^3 2u^2 du \right)$$

$$= \sin v \Big|_0^{2\pi} \cdot \frac{2}{3} u^3 \Big|_0^3 = 0$$

PROB 2)

$$\vec{F} = [y^2, x^2, z+x]$$

$$\text{curl } \vec{F} = [0, -1, 2x-2y]$$

Note: $\text{curl } \vec{F} \cdot \vec{n} = 2x-2y$ (Not constant)

Lets parameterize S in the $z=0$ plane

$$x = x \quad u = x$$

$$y = y$$

$$z = 0$$

$$u = x$$

$$\longrightarrow$$

$$v = y$$

$$\vec{r}(u,v) = [u, v, 0]$$

$$0 \leq u \leq 1$$

$$0 \leq v \leq u$$

} Region R

$$\vec{n} = [0, 0, 1]$$

$$\vec{r}_u = [1, 0, 0]$$

$$\vec{r}_v = [0, 1, 0]$$

$$\Rightarrow \vec{N} = \vec{r}_u \times \vec{r}_v = [0, 0, 1]$$

NOTE: UNDER this parameterization
 $\vec{N}$ is the same as $\vec{n}$

$$\Rightarrow \oint_C \vec{F} \cdot d\vec{r} = \iint_R (2u-2v) du dv = \int_0^1 \int_0^u (2u-2v) dv du$$

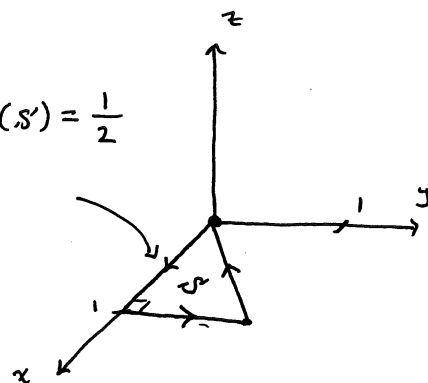
$$= \int_0^1 2uv - v^2 \Big|_0^u du$$

$$= \int_0^1 2u^2 - u^2 du$$

$$= \int_0^1 u^2 du$$

$$= \frac{u^3}{3} \Big|_0^1 = \frac{1}{3}$$

$$\text{Area}(S) = \frac{1}{2}$$



PROB 3]

$$\vec{F} = [0, z^3, 0]$$

$$\begin{aligned}\text{curl } \vec{F} &= [0 - 3z^2, 0 - 0, 0 - 0] \\ &= [-3z^2, 0, 0]\end{aligned}$$

Parameterizing S we have:

(Use cylindrical coordinates)

$$x = \cos \theta$$

$$y = \sin \theta$$

$$z = z$$

$$\begin{aligned}u &= \theta \\ v &= z\end{aligned}$$

$$\vec{r}(u, v) = [\cos u, \sin u, v]$$

$$\left. \begin{aligned}0 \leq u &\leq \frac{\pi}{2} \\ 0 \leq v &\leq 1\end{aligned} \right\} \text{R-Region}$$

$$\vec{r}_u = [-\sin u, \cos u, 0]$$

$$\vec{r}_v = [0, 0, 1]$$

$$\Rightarrow \vec{N} = \vec{r}_u \times \vec{r}_v = [\cos u, \sin u, 0]$$

By Stokes' Thm:

$$\oint_C \vec{F} \cdot d\vec{r} = \iint_R \text{curl}(\vec{F}(\vec{r})) \cdot \vec{N} \, du \, dv$$

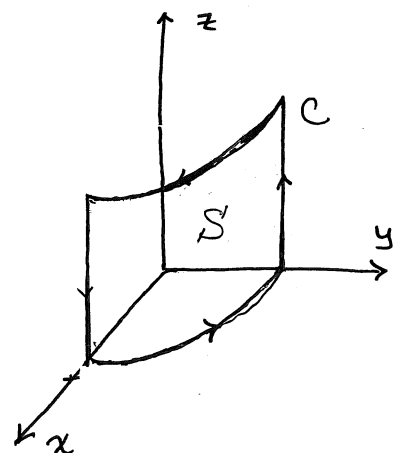
$$= \int_0^{\pi/2} \int_0^1 -3v^2 \cos u \, dv \, du$$

$$= \left(\int_0^{\pi/2} \cos u \, du \right) \left(\int_0^1 -3v^2 \, dv \right)$$

$$= \left(\sin u \Big|_0^{\pi/2} \right) \cdot \left(-v^3 \Big|_0^1 \right)$$

$$= (1 - 0)(-1 + 0)$$

$$= -1$$



PROB 4

$$z_1 = 5+i, \quad z_2 = 2-4i, \quad z_3 = a+bi$$

$$(a) \quad z_1 + z_2 = 7-3i$$

$$(b) \quad z_1 z_3 = (5+i)(a+bi) = 5a - b + (5b+a)i$$

$$(c) \quad \frac{z_2}{z_1} = \frac{2-4i}{5+i} \cdot \frac{5-i}{5-i} = \frac{10-4-2i-20i}{25+1} = \frac{6}{26} - \frac{22}{26}i$$

$$(d) \quad |z_2| = |2-4i| = \sqrt{4+16} = \sqrt{20}$$

$$(e) \quad z_1 - iz_2 = 5+i - 2i + 4i^2 = 1-i$$

$$(f) \quad \sqrt{-10} \cdot \sqrt{-6} = \sqrt{10}i \cdot \sqrt{6}i = \sqrt{60}i^2 = -\sqrt{60}$$

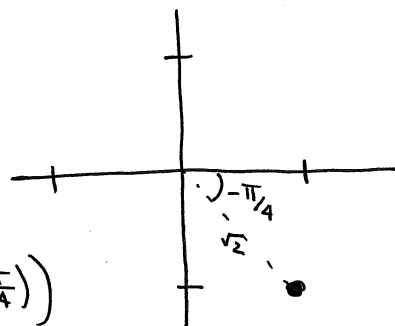
PROB 5

$$z_1 = 1 - i, \quad z_2 = 3i$$

$$(a) \quad r = |z_1| = \sqrt{1+1} = \sqrt{2}$$

$$\theta = -\frac{\pi}{4}$$

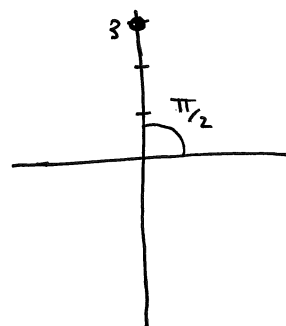
$$z_1 = \sqrt{2} e^{-\frac{\pi}{4}i} \quad \text{or} \quad \sqrt{2} \left(\cos\left(-\frac{\pi}{4}\right) + i \sin\left(-\frac{\pi}{4}\right) \right)$$



$$(b) \quad r = |z_2| = \sqrt{9+0} = 3$$

$$\theta = \frac{\pi}{2}$$

$$z_2 = 3 e^{\frac{\pi}{2}i} \quad \text{or} \quad 3 \left(\cos\left(\frac{\pi}{2}\right) + i \sin\left(\frac{\pi}{2}\right) \right)$$



$$(c) \quad \text{since } z_1 = \sqrt{2} e^{-\frac{\pi}{4}i}$$

$$\begin{aligned} \Rightarrow z_1^{12} &= (\sqrt{2})^{12} e^{(-\frac{\pi}{4}i)12} = 2^6 e^{-3\pi i} \\ &= 64 (\cos(-3\pi) + i \sin(-3\pi)) \\ &= -64 \end{aligned}$$

$$(d) \quad \text{since } z_2 = 3 e^{\frac{\pi}{2}i}$$

$$\Rightarrow z_2^{1/3} = 3^{1/3} e^{\frac{\frac{\pi}{2} + 2\pi k}{3}i}$$

$$k = 0, 1, 2$$

$$k=0: \quad = 3^{1/3} e^{\frac{\pi}{6}i}$$

$$k=1: \quad = 3^{1/3} e^{\frac{5\pi}{6}i}$$

$$k=2: \quad = 3^{1/3} e^{\frac{7\pi}{6}i}$$

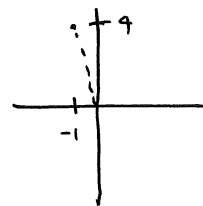
$$(d) \quad z^4 + z_1 = z_2$$

$$z^4 = z_2 - z_1 = -1 + 4i$$

$$z = (-1 + 4i)^{1/4} \longrightarrow$$

$$r = \sqrt{1+16} = \sqrt{17}$$

$$\theta = \pi - \tan^{-1}(4)$$



$$\Rightarrow z = (\sqrt{17})^{1/4} e^{\frac{\theta + 2\pi k}{4}}$$

$$k = 0, 1, 2, 3$$

$$k=0: = (17)^{1/8} e^{\frac{\theta}{4}}$$

$$k=1: = (17)^{1/8} e^{\frac{\theta + 2\pi}{4}}$$

$$k=2: = (17)^{1/8} e^{\frac{\theta + 4\pi}{4}}$$

$$k=3: = (17)^{1/8} e^{\frac{\theta + 6\pi}{4}}$$